arguments, or the side endorsed by the more prestigious source will be selected. However, if people have the necessary motivation and ability to evaluate the issue in a relatively objective manner, then the side presenting the highest quality arguments will be favored. If both sides present equally compelling positions, people may be motivated to resolve the ambiguity by using the available information to bolster their initial positions. Also, if one's previous knowledge on a issue is biased in favor of one's initial position, as will often be the case, then people will likely have greater ability to cognitively bolster their side but counterargue the opposing side. This process can lead to attitude polarization (e.g., Lord, Ross, & Lepper, 1979; see Chapter 5).

Retrospective

In this chapter we have examined and addressed some of the complications that arise in the Elaboration Likelihood Model. Among the complications are that variables can have one effect on elaboration when they are paired with some variables, but have different, even opposite, effects on elaboration when they are paired with other variables. Thus, some variables enhance elaboration at one level of a factor, but disrupt elaboration at another level of the same factor. Additionally, some variables affect information processing in a relatively objective manner when combined with some factors, but affect information processing in a more biased fashion when combined with other factors. Finally, we noted that the same factor can serve in any one of the three major roles postulated for variables by the ELM. That is, any one factor (e.g., source attractiveness) can serve as a persuasive argument in one context, act as a peripheral cue in another, and affect message elaboration in still a third situation. Fortunately, the ELM provides guidance as to when each of these processes is likely. In the final chapter, we offer some concluding remarks about the ELM.

Chapter 9

Epilogue

Introduction

We began this monograph with questions about the likely effectiveness of televised speeches given on New Year's Day by the American president and the Soviet premier to national audiences in each other's countries. The research and theory presented in this volume have indicated that communications such as these can produce persuasion via two fundamentally different routes. One route is based on the thoughtful (though sometimes biased) consideration of arguments perceived central to the merits of the issue under consideration, whereas the other is based on affective associations or simple inferences tied to peripheral cues in the persuasion context. When variables in the persuasion situation render the elaboration likelihood high, the first kind of persuasion occurs (central route). When variables in the persuasion situation render the elaboration likelihood low, the second kind of persuasion occurs (peripheral route). Importantly, there are different consequences of the two routes to persuasion. Attitude changes via the central route appear to be more persistent, resistant, and predictive of behavior than changes induced via the peripheral route (see Figure 1-1).

In outling the antecedents and consequences of the two routes to persuasion, the Elaboration Likelihood Model provides a framework from which to analyze the speeches of the U.S. and Soviet leaders. For example, a fundamental question is: How much thinking were the audiences likely to be doing about the communications? First, recall that the American president appeared during a news program in the Soviet Union, whereas the Russian premier appeared during televised coverage of a holiday parade in the United States. Thus, the elaboration likelihood was probably higher for the former than the latter speech. Because of this, the American leader may have been more likely to be judged on the quality of his arguments, whereas peripheral cues may have had a greater impact on attitudes toward the

Soviet leader's speech. Of course, this analysis is a gross oversimplification, and even if it was accurate, it would only be a starting point in the analysis. As we documented elsewhere in this monograph, a large number of factors must be considered in assessing the impact of the speeches (e.g., Did the President present strong or weak arguments? Was the processing of his speech objective or biased? Were the cues surrounding the Premier's speech positive or negative?). Despite these complexities, the ELM provides a unique and focusing perspective from which to analyze the situation. In this brief epilogue chapter, we will first discuss the elaboration likelihood concept in somewhat greater detail, and then we will consider the integrative potential of ELM.

Determinants of Elaboration Likelihood

In the preceding chapters we surveyed a wide variety of variables that proved instrumental in affecting the likelihood that a person would elaborate a persuasive communication, and thus the route to persuasion. In fact, one of the basic postulates of the ELM, that variables may affect persuasion by increasing or decreasing issue-relevant thinking proved useful in accounting for the effects of a seemingly diverse list of variables (see Chapters 3–5). The effects of these variables had been explained with many different theoretical accounts in the accumulated persuasion literature. The ELM was successful in tying the effects of these variables to one underlying process. We have also seen that many different variables could serve as peripheral cues, affecting persuasion without issue-relevant thinking. Again, the operation of a diverse list of variables is integrated by the common antecedents and consequents of their operation (see Chapters 6 and 7). Finally, we have seen that some variables were capable of serving in multiple roles, enhancing or reducing thinking in some contexts, and serving as simple acceptance or rejection cues in others (see Chapter 8).

In Figure 9-1 we present a general framework for thinking about the determinants of argument processing.[1] As we explained previously in Chapter 1, the likelihood that elaboration occurs can be viewed as being a function of the separable elements of motivation and ability.[2] By *motivation*, we mean the factor that propels and guides people's information processing and gives it its purposive character. We have seen that there are many

[1]The important influence of Heider (1958) in this figure should be apparent.

[2]Luck might also be considered a contributing factor. Luck refers to the transient and random circumstance in which an individual's ongoing ruminations bear upon an unanticipated attitude object or issue. As such, luck is not a particularly important determinant of elaboration likelihood. Hence, this factor is not considered further.

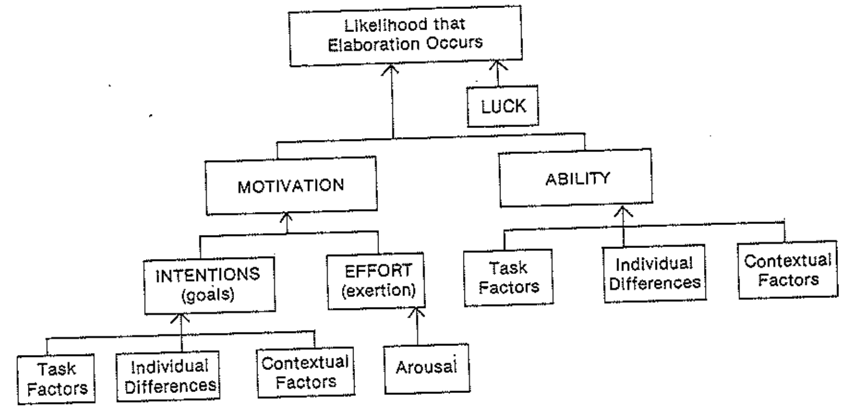

Figure 9-1. Schematic depiction of factors affecting the likelihood of elaboration.

variables that can affect a person's motivation to elaborate upon the content of a message. For instance, individuals have been found to be more motivated to think about the issue and message arguments when they are: (a) the only one rather than one of many assigned to evaluate the recommendation (e.g., Petty, Harkins, & Williams, 1980); (b) personally affected rather than unaffected by the adoption of the recommendation (e.g., Petty & Cacioppo, 1979b); (c) exposed to multiple sources rather than a single source espousing the virtues of a recommendation (e.g., Harkins & Petty, 1981a); and (d) high rather than low dispositionally in need for cognition (e.g., Cacioppo, Petty, & Morris, 1983). What these variables (and others affecting motivation) seem to have in common is that they act upon a directive, goal-oriented component which might be termed *intention*, and a nondirective, energizing information processing component which might be termed *effort* or exertion. As we have seen, many variables can influence a person's intention to think about the message arguments or issue. The *task variables* of personal relevance and responsibility, the *individual difference variable* of need for cognition, and the *contextual variable* of number of sources all influence an individual's intention to think about the persuasive communication presented.[3]

Intention is not sufficient for motivation, however, since one can want to think about a message or issue, but not exert the necessary effort to move from "intending" to "trying." For example, a person who is dispositionally

[3]We do not mean to suggest that these "intentions" are always conscious or deliberate in the sense that a person carefully plans to process the message. Instead, the goal of elaborating the message may be invoked quite spontaneously and naturally in certain situations or in certain people (cf., Sherman, in press).

Table 9-1. Factors Affecting Motivation and Ability to Process a Persuasive Communication in a Relatively Objective and Biased Manner

	Motivational Factors	Ability Factors
Relatively Objective Processing	Personal relevance (Petty & Cacioppo, 1979b)	External distraction (Petty, Wells, & Brock, 1976)
	Personal responsibility (Petty, Harkins, & Williams, 1980)	Transient heart rate (Cacioppo, 1979)
		Recipient Posture (Petty, Wells, Heesacker, Brock, & Cacioppo, 1983)
	Number of sources (Harkins & Petty, 1981a)	
	Need for cognition (Cacioppo, Petty, & Morris, 1983)	Moderate message repetition (Cacioppo & Petty, 1985)
Relatively Biased Processing	Forewarning of message content (Petty & Cacioppo, 1977)	Self-schemata (Cacioppo, Petty, & Sidera, 1982)
	Forewarning of persuasive intent (Petty & Cacioppo, 1979a)	Hemispheric asymmetry (Cacioppo, Petty, & Quintanar, 1982)
	Excessive message repetition (Cacioppo & Petty, 1979b)	Vertical/horizontal head movements (Wells & Petty, 1980)

high in need for cognition may have the general goal of thinking, but may not exert the effort on a particular day due to mental fatigue. Little research has been directed explicitly at specifying what variables affect this stage, although effort has been linked previously to drive and arousal (e.g., Kahneman, 1973). The data obtained using measures of autonomic and skelotomotor activation have not provided clear support for this connection, however, and interesting questions remain regarding exactly which variables act on cognitive effort and whether more sophisticated conceptualizations and measures of arousal will yield evidence for the hypothetical link outlined in Figure 9-1 (e.g., see Cacioppo & Petty, in press).

If both intention and effort are present, then motivation to think about the advocacy will exist. Message elaboration, however, also requires that the individual have the *ability* to process the message. As we have seen elsewhere in this monograph, there are a variety of variables that can affect elaboration ability. *Task variables* such as message comprehensibility (e.g., Eagly, 1974), *individual difference variables* such as intelligence (e.g., Eagly & Warren, 1976) and relative hemispheric activation (e.g., Cacioppo, Petty, & Quintanar, 1982), and *contextual variables* such as distraction (e.g., Petty, Wells, & Brock, 1976) and moderate message repetition (e.g., Cacioppo & Petty, 1979b) all influence an individual's ability to think about message

arguments. It can be seen that all factors affecting the person's *opportunity* to process message arguments, such as external distraction, would fall under the headings of either task or contextual factors.

Figure 9-1 does not address what happens next, but rather the ELM as outlined in Figure 1-1 does. If task, individual, and contextual factors in the persuasion situation combine to render motivation and ability to process high, then the arguments in the persuasive communication will be elaborated. The nature of this elaboration will be determined by whether the motivational and ability factors combine to yield relatively objective or relatively biased processing. If the processing is relatively objective, elaboration will be governed mostly by the subjective cogency of the issue-relevant arguments contained in the message. If the processing is relatively biased, issue-relevant thoughts will be governed more by factors such as the person's initial attitude schema. Table 9-1 summarizes some of the major variables examined in our research that affect motivation and ability to elaborate in a relatively objective or biased manner.

Integrative Potential of the ELM

We noted early in this monograph that reviewers of the attitude change literature have been disappointed by the fact that many conflicting effects have been observed even for ostensibly simple variables. For example, manipulations of source expertise have sometimes increased persuasion, sometimes had no effect, and sometimes decreased persuasion. Similarly, studies testing different theories of persuasion have sometimes found the theory to be useful in predicting attitude change, and other times have found the theory to be unpredictive. For example, self-perception processes appear to operate under some conditions, but not others.

The Elaboration Likelihood Model represents an attempt to place these many conflicting results and theories under one conceptual umbrella by specifying the major processes underlying persuasion and indicating how many of the traditionally studied variables and theories relate to these basic processes. Thus, we have seen that a seemingly simple variable like source credibility is actually capable of affecting persuasion in rather complex ways. The ELM, however, elucidates the general conditions under which these different effects are likely to operate. Specifically, we have seen that source expertise acts as a simple peripheral cue when the elaboration likelihood is very low, but not when it is very high (e.g., Petty, Cacioppo, & Goldman, 1981). When the elaboration likelihood is moderate, source expertise affects a person's motivation to think about the issue-relevant arguments provided (e.g., Heesacker, Petty, & Cacioppo, 1983).

Similarly, the ELM may prove useful in specifying the domains of different attitude theories. Theories proposing that attitudes change because of simple affective cues (e.g., Staats & Staats, 1957, 1958; Kelman, 1958), or cognitive heuristics (e.g., Bem, 1972; Chaiken, in press) should operate mostly when the elaboration likelihood is low. On the other hand, persuasion models emphasizing issue-relevant cognitive activity (e.g, Ajzen & Fishbein, 1980; McGuire, 1964; Petty, Ostrom, & Brock, 1981) should operate mostly when the elaboration likelihood is high. Consider the decade-long controversy regarding the viability of dissonance and self-perception theories. Both conceptualizations predicted the same pattern of attitude results in many situations, though for different reasons (Greenwald, 1975). Self-perception theory postulated that attitude change resulted from a simple inference based on the observation of one's own behavior (Bem, 1972). Dissonance theory, on the other hand, postulated a much more cognitively active process. As a result of dissonance, people were believed to engage in rationalizing thoughts that would justify their attitude-inconsistent action (i.e., biased elaboration; Festinger, 1957). In the language of the ELM, self-perception proposes a peripheral route to persuasion, whereas dissonance is more central. Thus, dissonance processes should be more likely to operate when the elaboration likelihood is relatively high (i.e., conditions of high issue relevance, consequences, personal responsibility, prior knowledge, etc.) rather than low. Research is generally supportive of this view (e.g., Fazio, Zanna, & Cooper, 1977; Cooper & Fazio, 1984). On the other hand, self-perception processes should be more likely to operate when the elaboration likelihood is relatively low (i.e., conditions of low issue relevance, consequences, responsibility, knowledge, etc.) rather than high. Research also supports this proposition (e.g., Chaiken & Baldwin, 1981; Taylor, 1975; Wood, 1982).[4] If our analysis is correct, then a number of interesting (and so far untested) consequences follow. For example, if Person A becomes more favorable toward some behavior because of a dissonance producing experience, and Person B become equally more favorable toward the same behavior because of an instance of self-perception, the attitude of Person A should show greater resistance to counterpersuasion (see Chapter 7).

In proposing that certain variables can increase persuasion in some situations but decrease persuasion in others, and that some theories operate in some situations but not others, the ELM is broadly consistent with McGuire's (1983) "contextualist" epistemology for social psychology. In short, the ELM proposes that many of the effects and theories uncovered by persuasion researchers over the years are valid, but their domain of operation is limited (Petty & Cacioppo, 1981a). Importantly, the ELM specifies two general categories of persuasion processes (central and peripheral) and elucidates their antecedents and consequences. In doing this, the ELM suggests *when* many of the traditionally studied variables will

[4]Of course, even self-perception requires some *minimal* thought sufficient for an inference to be made (e.g., Scott & Yalch, 1978). The same applies to the operation of many other peripheral cues.

work and *how* they will work. It also helps to specify the domains of many of the extant mini-theories of persuasion.

Finally, we note that although we have confined our discussion (and our own research) in this monograph mostly to the traditional influence situation in which a persuasive communication on one side of an issue (e.g., raising tuition) is presented to a message recipient, we believe that the ELM also has more general applicability to other topics studied by social psychologists. For example, in research on majority versus minority influence, subjects do not hear one message but rather receive conflicting viewpoints from both a majority and a minority group (e.g., Moscovici, 1980). Researchers in this area have suggested that under certain conditions, minorities induce influence via the central route whereas the influence of majorities is more peripheral (Maass & Clark, 1984; Nemeth, 1986). Interpersonal attraction might be governed by both central and peripheral processes as well. For example, early in a relationship when prior knowledge and involvement are relatively low, peripheral physical cues such as attractiveness (e.g., Berscheid & Walster, 1974), or contextual cues such as room temperature (e.g., Griffit & Vietch, 1971) may be influential. As prior knowledge and involvement increase, these cues should become less important as the central merits of the person are processed.[5] Likewise, attitudes toward oneself might be similarly influenced. Specifically, there may be both central and peripheral aspects of the self-concept with the first being formed because of extensive processing of self-relevant information and the second resulting from relatively minimal processing of various cues. In addition, not only may attitudes and beliefs be affected via the central and peripheral routes, but so too may other judgments such as causal attributions (see discussion of Borgida & Howard-Pitney, 1983, in Chapter 6).

Conclusions

Perhaps the greatest strength of the Elaboration Likelihood Model is that it specifies the major ways in which variables can have an impact on persuasion, and it points to the major consequences of these different

[5]Importantly, as we noted in Chapter 1, the dimensions that are central to the merits of an attitude object can vary from person to person (e.g., Cacioppo, Petty, & Sidera, 1982; Snyder & DeBono, 1985) and from situation to situation. Thus, for some people, attractiveness may be a central merit (employed when personal involvement is relatively high) whereas intelligence is a peripheral cue (used in evaluation when personal involvement is relatively low). For other people, the opposite may hold. In addition, different situations may render different attributes as central. For example, in judging a person's prospects for graduate school, intelligence is central but attractiveness is peripheral. In judging a person for a modeling career, the opposite may hold.

mediational processes. In one sense, the ELM is rather simple. It indicates that variables can affect persuasion in a limited number of ways: a variable can serve as a persuasive argument, serve as a peripheral cue, or affect argument scrutiny in either a relatively objective or a relatively biased manner. In confining the mediational processes of persuasion to just these possibilities, the ELM provides a unique simplifying and organizing framework that may be applied to many of the traditionally studied source, message, recipient, and context variables. The postulates of the ELM in its present state of development do *not* ultimately indicate *why* certain arguments are strong or weak, why certain variables serve as cues, or why certain variables affect information processing. Instead, the ELM limits the mediational processes of persuasion to a finite set and specifies, in a general way at least, the conditions under which each mediational process is likely to occur and the consequences of these processes. In doing this, the ELM may prove useful in providing a guiding set of postulates from which to interpret previous work, and in posing an interesting set of hypotheses to be explored in future studies.

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